① Processos Estocásticos (Dia 26/10/2026)

Recapitulando:

$\varphi: \Omega \times [0,T] \to \mathbb{R}$ é simples (i.e, $\varphi \in \mathcal{H}_0^2[0,T]$) se

$$\varphi_t = \sum_{i=0}^{n-1} a_i \mathbb{1}_{(t_i, t_{i+1}]}(t)$$

$\mathbb{E}(a_i^2) < \infty$, $0 = t_0 < t_1 < \cdots < t_n = T$.

$$\mathcal{H}^2[0,T] = \left\{ \phi \in \mathcal{M} : \mathbb{E}\left[\int_0^T \phi_t^2 \, dt\right] < \infty \right\}$$

($\mathcal{M}$: mensuráveis, adaptados.)

Definimos $I : \mathcal{H}_0^2[0,T] \longrightarrow L^2(\Omega, \mathcal{F}, \mathbb{P})$

$$I(\varphi) = \sum_{i=0}^{n-1} a_i (B_{t_{i+1}} - B_{t_i}) \quad \forall \varphi \in \mathcal{H}_0^2[0,T].$$

- $\mathcal{H}_0^2[0,T] \subseteq \mathcal{H}^2[0,T]$ (é denso aqui)
- $\mathcal{H}_0^2$ é espaço de Hilbert.
- $\|I\|_{\mathcal{H}^2} = 1$ (mais ainda, $\|I(\varphi)\|_{L^2} = \|\varphi\|_{\mathcal{H}^2} \ \forall \varphi \in \mathcal{H}_0^2$)

$\Rightarrow$ Podemos extender I para $\mathcal{H}^2$, de maneira contínua:

Tome $\varphi \in \mathcal{H}^2$ e seja $(\varphi^{(n)})_{n \in \mathbb{N}} \subseteq \mathcal{H}_0^2$ t.q. $\|\varphi^{(n)} - \varphi\|_{\mathcal{H}^2} \to 0$.

$$I(\varphi) := L^2\text{-}\lim I(\varphi^{(n)})$$

(2) Em particular, $\|I(\varphi)\|_{L^2} = \|\varphi\|_{\mathcal{H}^2} \quad \forall \varphi \in \mathcal{H}^2$ (Isometria de Itô)

Exemplo: é falso que "$I(B) = \int_0^T B_t \, dB_t = \frac{B_T^2}{2}$".

Isso seria verdade se B tivesse variação total finita.

- Seja $\varphi_t = B_t$. $\Rightarrow$ $\varphi \in \mathcal{H}^2$. De fato, $B_t \in \mathcal{F}_t$,

$$\int_0^T \mathbb{E}(B_t^2)\, dt = \int_0^T t \, dt = T^2/2 < \infty.$$

- Seja $t_i = \frac{i}{n} \cdot T$, $\varphi_t^{(n)} = \sum_{i=0}^{n-1} B_{t_i} \mathbb{1}_{(t_i, t_{i+1}]}(t)$.

- Queremos $\|\varphi^{(n)} - \varphi\|_{\mathcal{H}^2} \to 0$.

De fato, $(\varphi_t - \varphi_t^{(n)})^2 = \left(B_t - \sum_{i=0}^{n-1} B_{t_i} \mathbb{1}_{(t_i, t_{i+1}]}(t)\right)^2$

$$= \left(B_t \sum_{i=0}^{n-1} \mathbb{1}_{(t_i, t_{i+1}]}(t) - \sum_{i=0}^{n-1} B_{t_i} \mathbb{1}_{(t_i, t_{i+1}]}(t)\right)^2$$

$$= \left(\sum_{i=0}^{n-1} (B_t - B_{t_i}) \mathbb{1}_{(t_i, t_{i+1}]}(t)\right)^2$$

$$= \sum_{i=0}^{n-1} (B_t - B_{t_i})^2 \mathbb{1}_{(t_i, t_{i+1}]}(t)$$

(2)

$$\|\varphi - \varphi^{(n)}\|_{\mathcal{H}^2} = \int_0^T \mathbb{E}\left[(\varphi_t - \varphi_t^{(n)})^2\right] dt$$

$$= \int_0^T \mathbb{E}\left[\sum_{i=0}^{n-1} (B_t - B_{t_i})^2 \mathbb{1}_{(t_i, t_{i+1}]}(t)\right] dt$$

$$= \sum_{i=0}^{n-1} \int_0^T \underbrace{\mathbb{E}\left[(B_t - B_{t_i})^2\right]}_{(t - t_i)} \mathbb{1}_{(t_i, t_{i+1}]}(t)\, dt$$

$$= \sum_{i=0}^{n-1} \underbrace{\frac{(t_{i+1} - t_i)}{2}}_{T^2/2n^2} = \frac{T^2}{2n} \longrightarrow 0$$

- $I(\varphi^{(n)}) = \sum_{i=0}^{n-1} B_{t_i}(B_{t_{i+1}} - B_{t_i})$

- $B_{t_i}(B_{t_{i+1}} - B_{t_i}) =$ [illegible]

$$= \left(\frac{1}{2}B_{t_{i+1}} + \frac{1}{2}B_{t_i} + \frac{1}{2}B_{t_i} - \frac{1}{2}B_{t_{i+1}}\right)\left(B_{t_{i+1}} - B_{t_i}\right)$$

$$= \frac{1}{2}\left(B_{t_{i+1}}^2 - B_{t_i}^2\right) - \frac{1}{2}\left(B_{t_{i+1}} - B_{t_i}\right)^2$$

$$\Rightarrow I(\varphi^{(n)}) = \underbrace{\frac{1}{2}\sum_{i=0}^{n-1}\left[B_{t_{i+1}}^2 - B_{t_i}^2\right]}_{= B_T^2 - B_0^2} - \underbrace{\frac{1}{2}\sum \left(B_{t_{i+1}} - B_{t_i}\right)^2}_{\to V_2[0,T] = T}$$

(4)

$\Rightarrow$ [illegible]

$\Rightarrow I(\varphi^{(n)}) \longrightarrow \frac{1}{2}B_T^2 - \frac{1}{2}T$

$\Rightarrow I(\varphi) = \frac{1}{2}B_T^2 - \frac{1}{2}T$

Integral de Itô como um martingal

Queremos $\int_0^t \varphi_t dB_t = I(\mathbb{1}_{[0,t]}\varphi) \quad \forall t \in [0,T]$

$\exists \Omega_t$ t.q. $I(\mathbb{1}_{[0,t]}\varphi)(\omega)$ está definido $\forall \omega \in \Omega_t$.

Porém, é possível que $\mathbb{P}\left(\bigcap_{t\in[0,T]} \Omega_t\right) < 1$.

Teo(*) Dado $\varphi \in \mathcal{H}^2[0,T]$, $\exists X = (X_t)_{t\in[0,T]}$ martingal contínuo c.r.a $(\mathcal{F}_t^B)_{t\geq 0}$ t.q. $\mathbb{P}\left(X_t = I(\varphi \mathbb{1}_{[0,t]})\right) = 1 \quad \forall t \in [0,T]$.

Teo (Desig. maximal de Doob): $(M_t)_{t\geq 0}$ sub martingal contínuo não negativo c.r.a $(\mathcal{F}_t)_{t\geq 0}$. Então

$$\lambda^p \mathbb{P}\left(\sup_{t\in[0,T]} M_t > \lambda\right) \leq \mathbb{E}(M_T^p)$$

$\forall \lambda > 0, \forall p \geq 1.$

(5)

Prova de (*)

Dado $\varphi \in \mathcal{H}^2$, tome $(\varphi^{(n)})_{n\in\mathbb{N}} \subseteq \mathcal{H}_0^2$ t.q $\varphi^{(n)} \xrightarrow{\mathcal{H}^2} \varphi$.

$$\Rightarrow \varphi^{(n)} \mathbb{1}_{[0,t]} \xrightarrow{\mathcal{H}^2} \varphi \mathbb{1}_{[0,t]}.$$

$$\Rightarrow I(\varphi^{(n)} \mathbb{1}_{[0,t]}) \xrightarrow{L^2} I(\varphi \mathbb{1}_{[0,t]})$$

Defina $X_t^{(n)} = I(\varphi^{(n)} \mathbb{1}_{[0,t]})$

$$\varphi_s^{(n)} = \sum_{i=0}^{m_n - 1} a_i^{(n)} \mathbb{1}_{(t_i^{(n)}, t_{i+1}^{(n)}]}(s)$$

$\exists\, k: \quad t_k^{(n)} < t \leq t_{k+1}^{(n)}$,

$$\Rightarrow \varphi_s^{(n)} \mathbb{1}_{[0,t]}(s) = a_k^{(n)} \mathbb{1}_{(t_k^{(n)}, t]}(s) + \sum_{i=0}^{k-1} a_i^{(n)} \mathbb{1}_{(t_i^{(n)}, t_{i+1}^{(n)}]}$$

$$\Rightarrow X_t^{(n)} = I(\varphi^{(n)} \mathbb{1}_{[0,t]}) = a_k^{(n)} (B_t - B_{t_k^{(n)}}) + \sum_{i=0}^{k-1} a_i^{(n)} (B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}})$$

$\Rightarrow (X_t^{(n)})_{t\in[0,T]}$ é contínua, martingal, adaptado a $(\mathcal{F}_t^B)_{t\geq 0}$.

$\Rightarrow X^{(n)} - X^{(m)}$ é

$\rightarrow$ Desig max de Doob $\Rightarrow$ $\mathbb{P}\left(\sup_{0\leq t\leq T} |X_t^{(n)} - X_t^{(m)}| \geq \varepsilon \right)$

$$\leq \frac{1}{\varepsilon^2} \mathbb{E}\left[|X_T^{(n)} - X_T^{(m)}|^2 \right] = \frac{1}{\varepsilon^2} \mathbb{E}\left[|I(\varphi^{(n)}) - I(\varphi^{(m)})|^2 \right]$$

...